WESLEY JANES

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EDUCATION & HONORS

Mississippi State University, Bagley College of Engineering **Starkville, MS**
M.S. Aerospace Engineering, *Thesis Approach* *Expected Graduation: May 2028*
B.S. Aerospace Engineering, *Concentration in Astronautics* | **GPA: 3.44 / 4.0** *August 2021 – May 2026*
Shackouls Honors College | President's List / Dean's List | Eagle Scout

Relevant Coursework: Aerothermodynamics, Aerospace Controls, Spacecraft Attitude Dynamics, Aerostructure Design, Orbital Mechanics, Fluid Mechanics, Spacecraft Design, Vibrations, Compressible Aerodynamics, Spacecraft Propulsion, Circuits, Computer Programming, Calculus, Physics.

ENGINEERING EXPERIENCE

Applied Aerodynamics Research at Mississippi State University **Starkville, MS**
Undergraduate Research Assistant & Co-Author *August 2024 – Present*
• Validated aerodynamic repeatability of modified leading-edge airfoils by **executing subsonic wind tunnel tests**, correlating experimental data with **XFOIL** predictions to **ensure ±5% data fidelity**.
• Spearheaded **oil flow visualization studies** across multiple airfoil profiles to identify boundary surface flow behaviors, co-authoring a technical paper for **journal publication (expected publication: June 2026)**.

Xipiter Unmanned Aerial Systems Club **Starkville, MS**
UAS 2025-2026 Design – CAD & Software Team Lead *January 2025 – Present*
• **Directed a multidisciplinary team of 14 student-engineers** in the structural design of a custom UAS airframe in **SolidWorks**, optimizing internal structural components to **integrate avionics systems**.
• Developed mission-critical **Python** scripts to automate flight control parameters, directly enabling the aircraft to execute the designated competition mission profile.
• Supervised **composite airframe fabrication**, applying **GD&T** standards to guarantee precision integration between carbon fiber components and internal sheet metal mounts.

Systems & Mission Design Analysis of the X-15 Hypersonic Vehicle **Starkville, MS**
Modeling & Simulation Lead *August 2025 – December 2025*
• Translated historical **mission requirements** into technical specifications to develop a **SolidWorks** airframe model with **98%** geometric accuracy.
• Performed **3D thermal-structural analysis** using **ANSYS** with a accuracy window of **±5%**, simulating temperature contours at Mach 8 to evaluate engineers of the 1960's **design requirements** and performance limits of the thermal ablative coating.
• Presented technical findings to **nationally recognized museums**, delivering a mission briefing that correlated modern computational results with historical flight data for public exhibition.

Aerostructure Design of Stiffened Panels of Multiple Failure Modes **Starkville, MS**
Manufacturing Lead *August 2025 – December 2025*
• Engineered structural airframe models in **SolidWorks** and performed **Abaqus FEA** to identify critical stress concentrations, ensuring the assembly maintained a **1.5 factor of safety** under predicted failure loads.
• Optimized manufacturing process by applying **GD&T** principles to technical drawings and overseeing **sheet metal fabrication process**, ensuring precision cutting, bending, drilling, and riveting.
• Achieved **95%** accuracy in results between UTM data, computational FEA, and theoretical calculations.

Aerodynamic Analyses of Modified Leading Edges **Starkville, MS**
Aerodynamic Testing Lead *August 2024 – May 2025*
• Engineered **16 advanced airfoil geometries** using **SolidWorks** and resin-based **additive manufacturing**, reducing prototype post-processing times by **24+ hours** and enabling rapid iteration of leading-edge modifications.
• **Streamlined data collection** by conducting subsonic wind tunnel experiments and supporting analysis through **Monte Carlo** simulations with real-time **LabVIEW data acquisition**, ensuring high-fidelity aerodynamic profiling across multiple Reynolds numbers.
• Developed custom **MATLAB** post-processing scripts to automate lift, drag, and moment coefficients, cutting data reduction time by **75%**.

Compressible Flow Analysis of Rocket Nozzles **Starkville, MS**
Propulsion Analysis Lead *August 2024 – December 2024*
• Simulated converging/diverging nozzle performance in **ANSYS**, analyzing shockwave formations and choked flow to optimize contours at target altitudes; achieved a **90%** correlation between computational results and isentropic flow equations.
• Automated performance visualization via **MATLAB**, slashing data processing time by **80%** and enabling rapid iteration of mass flow calculations.

The Lilly Company **Memphis, TN**
IT Assistant *May 2023 – December 2023*
• Optimized system deployment workflows by utilizing **AI/ML** and standardizing hardware configuration, resulting in a **50%** increase in technical setup efficiency for fleet-wide industrial mobile assets.
• Managed critical network infrastructure and data integrity, resolving **200+** service tickets to ensure seamless system functionality for regional logistical operations.

TECHNICAL SKILLS

Software: MATLAB, ANSYS, SolidWorks, LabVIEW, AutoCAD, Abaqus, STK, XFOIL, Python, C, C++, Overleaf, Microsoft Suite, Adobe Suite.
Manufacturing: Additive Manufacturing, CNC Machining, Composite Fabrication, Sheet Metal Fabrication, GD&T.